

# MAHADEV

## Final-Year Electrical Engineering Student | Power Systems Renewable Energy | Embedded & IoT-Based Energy Systems

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### PROFESSIONAL PROFILE

Final-year, 8th-semester B.E. Electrical Engineering student at Mehran University of Engineering and Technology, Jamshoro, with practical experience in power systems, renewable energy, embedded systems, IoT monitoring, hardware prototyping, and industrial electrical operations. Completed internships at HESCO and JPCL and developed multiple contract-based engineering projects involving ESP32, Arduino, sensors, relay control, motor control, solar energy, and real-time web monitoring. Skilled in MATLAB/Simulink, ETAP, Proteus, Multisim, EdrawMax, Arduino IDE, embedded programming, circuit-diagram development, Veroboard fabrication, soldering, testing, debugging, and technical documentation. Currently completing a final-year research project on pressure-loss-constrained energy recovery from industrial exhaust airflow using a hybrid VAWT–PMSG system.

### EDUCATION

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|-------------------------------------------------------------------------------------|-------------|
| <b>Bachelor of Engineering (B.E.) in Electrical Engineering</b>                     | 2022 – 2026 |
| Mehran University of Engineering & Technology (MUET), Jamshoro<br>CGPA: 3.00 / 4.00 |             |
| <b>Intermediate (Pre Engineering)</b>                                               | 2022        |
| Govt. H/Sec School Nabisar Sahar                                                    |             |
| <b>Matriculation (Science)</b>                                                      | 2020        |
| Govt. H/Sec School Nabisar Sahar                                                    |             |

### EXPERIENCE

#### 132 kV Qasimabad Grid Station – HESCO

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| Power Systems Intern (4 Weeks)                                                                                                                                                                                                                                                                                                                                                                                              | July 2025 – Aug 2025 |
| <ul style="list-style-type: none"><li>Observed 132/11 kV grid station equipment, operation, and protection systems.</li><li>Studied power transformers, CTs, PTs, SF<sub>6</sub> breakers, relays, isolators, and busbars.</li><li>Reviewed single-line diagrams, feeder operation, load management, and switching procedures.</li><li>Prepared a technical internship report based on grid station observations.</li></ul> |                      |

#### Hyderabad Electric Supply Company (HESCO)

|                                                                                                                                                                                                                                                                                                                         |                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|
| Electrical Engineering Intern (2 Weeks)                                                                                                                                                                                                                                                                                 | Dec 2023 – Jan 2024 |
| <ul style="list-style-type: none"><li>Observed distribution system operation, feeder planning, and grid safety practices.</li><li>Studied conductor selection, protection coordination, and distribution management.</li><li>Gained field exposure in Chief Engineer P&amp;E and GSO HESCO Hyderabad offices.</li></ul> |                     |

#### Jamshoro Power Company Limited (JPCL – GENCO-1)

|                                                                                                                                                                                                                                                                                                                 |                       |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Engineering Intern (1 Month)                                                                                                                                                                                                                                                                                    | June 2024 – July 2024 |
| <ul style="list-style-type: none"><li>Observed thermal power plant operation and electrical support systems.</li><li>Studied plant safety procedures, maintenance practices, and troubleshooting methods.</li><li>Developed practical understanding of generation-side industrial electrical systems.</li></ul> |                       |

### PROJECTS

#### FINAL YEAR PROJECT (ONGOING)

**Utilization of Industrial Exhaust Air to Drive a Small-Scale Wind Turbine for Power Generation and Energy Recovery**

- Researching energy recovery from industrial exhaust airflow in textile environments.
- Studying duct-integrated wind turbine design, airflow behavior, and pressure drop.
- Working on PMSG generation, rectification, DC conversion, MPPT, and storage interface.
- Performing MATLAB/Simulink modelling and preparing technical system diagrams.
- Developing feasibility analysis, poster material, and conference-style documentation.

## **SELECTED ENGINEERING PROJECTS**

### **IoT-Based Smart Energy Management System with Theft Detection Using Dual Power Source Solar and WAPDA EEP32 | Solar + WAPDA | ATS | IoT**

Dec 2025

*Complete*

- Designed a dual-source energy system using solar/inverter and WAPDA grid supply.
- Integrated automatic transfer switching for uninterrupted source changeover.
- Monitored solar voltage, battery voltage, grid voltage, load current, and power.
- Developed theft detection using ACS712 sensing and abnormal current monitoring.
- Built a Wi-Fi-based dashboard for live source, alarm, and system-status monitoring.
- Integrated ESP32, ZMPT101B, ACS712, relays, LCD, inverter, battery, and solar panel.

### **Design and Fabrication of an Automatic Solar Panel Cleaning System**

Jan 2026

**Arduino UNO | BTS7960 (IBT-2) | RTC | Bluetooth | Relay Control**

*Complete*

- Fabricated a solar panel cleaning prototype using chain drive and wiper mechanism.
- Controlled bidirectional DC motor movement using Arduino UNO and BTS7960 driver.
- Developed Bluetooth manual control using HC-05 and scheduled control using DS3231 RTC.
- Integrated relay-based water pump control for synchronized cleaning operation.
- Used limit switches for safe automatic stopping at both ends of the panel.
- Tested wiring, motor movement, pump operation, and cleaning mechanism reliability.

### **IoT-Based LPG and Natural Gas Leakage Detection System with Automatic Servo-Actuated Flow Control**

**ESP32 | MQ-2 | MQ-5 | Flame Detection | Servo Control | IoT Dashboard Completed Contract-Based Project 2026**

- Developed an ESP32-based system for LPG, natural-gas, and flame detection with automatic safety control.
- Integrated MQ-2, MQ-5, flame sensor, servo valve, exhaust fan, buzzer, LEDs, LCD, RTC, and microSD logging.
- Implemented SAFE, WARNING, GAS LEAK, and FIRE modes with automatic valve shut-off and alarm response.
- Developed a local Wi-Fi dashboard for real-time monitoring and authorised manual control.
- Designed the EdrawMax circuit diagram and completed firmware, soldering, calibration, testing, and hardware integration.

### **Design and Fabrication of Smart Energy Meter with Priority Load Control**

**ESP32 | PZEM-004T | CT | Relay Control | Load Prioritization | Prepaid Logic**

*Ongoing Contract-Based Project*

- Developed a smart energy meter concept with automatic priority load control.
- Integrated voltage, current, power, energy, frequency, and power factor monitoring.
- Designed load-shedding logic to disconnect low-priority loads during overload.
- Used ESP32, PZEM-004T, CT, relay module, OLED/LCD, buzzer, and load circuit.
- Prepared block diagram, circuit diagram, and implementation workflow.

### **Smart Monitoring of Transmission Lines for Enhancing Power Grid Reliability**

**MATLAB/Simulink | Power Systems | Fault Detection | Sag | Temperature | Vibration**

*Ongoing Contract-Based Project*

- Developed a MATLAB/Simulink model for transmission-line condition monitoring.
- Simulated voltage, current, temperature, sag, vibration, overload, and fault cases.
- Designed threshold alarm logic for overcurrent, undervoltage, and overtemperature.
- Used RMS, per-unit voltage, loading index, thermal response, and sag estimation.

## **MINI ACADEMIC PROJECTS**

## Temperature Monitoring System Using Arduino and LM35

Apr 2025

### Arduino UNO | LM35 | Display Interface

- Developed a real-time temperature monitoring system using LM35 temperature sensor and Arduino UNO.
- Displayed temperature readings on a screen and strengthened practical skills in sensor interfacing and embedded monitoring.

## Traffic Light Controller (Digital & 555 Timer Based)

Nov 2024

- Designed a traffic light control system using 555 Timer IC and CD4017 decade counter
- Performed timing analysis and implemented sequential logic control for signal switching

## Fire Alarm System

Apr 2024

- Designed and implemented a fire alarm system using BC547 NPN transistors, IR receiver LEDs, buzzer, and indicator LEDs
- Gained hands-on experience in circuit design, component selection, and system testing

## Digital-to-Analog Converter (DAC) – R-2R Ladder

- Developed a 4-bit R-2R ladder DAC using op-amps (IC 741)
- Studied binary-to-analog signal conversion and output voltage scaling

## Star-Delta Connection (Three-Phase System)

Oct 2023

- Implemented star and delta configurations in a three-phase circuit using resistive loads
- Measured and analyzed line voltage, phase voltage, line current, and phase current
- Developed understanding of power distribution and motor starting techniques

## Charging and Discharging of Capacitor

Apr 2023

- Designed a circuit to analyze charging and discharging characteristics of capacitors through resistive networks
- Gained practical experience in RC time constant analysis and transient behavior

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## CERTIFICATIONS

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|--------------------------------------------------------------------------------|------|
| • MATLAB Onramp – MathWorks                                                    | 2025 |
| • HESCO Internship Certificate                                                 | 2025 |
| • HESCO Internship Certificate                                                 | 2024 |
| • JPCL / GENCO-I Internship Certificate                                        | 2024 |
| • IoT and Startup Culture Workshop – QS ImpACT Pakistan and Indus Techetronics | 2024 |
| • Condition-Based Maintenance of Equipment in Power Sector – IEEE PES          | 2024 |
| • Unlocking Scholarship Opportunities for Undergraduates – IEEE PES MUET       | 2024 |
| • Geothermal Energy Extraction and Infrastructure Planning – Technical Session | 2024 |
| • IEEE Day 2024: Research and Innovation with IoT & AI                         | 2024 |
| • Clean Energy and Climate Resilient Housing Workshop – RDF                    | 2026 |
| • Digital Marketing – DigiSkills                                               | 2024 |
| • Digital Skills: Artificial Intelligence – Accenture                          |      |

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## RESEARCH AND ACADEMIC PRESENTATION

### Poster Presenter

### 8th International Conference on Energy, Environment and Sustainable Development (EESD 2026)

MUET Jamshoro | 25–26 March 2026

- Presented a poster at the international conference organized by the Energy & Environmental Engineering Research Group (EEERG), MUET Jamshoro.
- Focused on energy, environment, sustainable development, and applied engineering research.

## PROFESSIONAL MEMBERSHIPS

IEEE (Institute of Electrical and Electronics Engineers)

(2024 – Present)

- IEEE Student Member (Karachi Section)
  - IEEE Power & Energy Society (PES)
  - IEEE Circuits & Systems Society
  - IEEE Industry Applications Society
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## TECHNICAL SKILLS

### Power and Energy Systems:

Power systems, renewable energy systems, smart grids, solar energy systems, power generation, grid station operation, transmission and distribution systems, load management, energy recovery systems.

### Simulation and Engineering Software:

MATLAB, Simulink, ETAP 19.0.1, Proteus, Multisim, Arduino IDE, Wondershare EdrawMax,

### Embedded Systems and Hardware:

Arduino UNO, ESP32, sensors, relay modules, ACS712, ZMPT101B, LM35, DS3231 RTC, HC-05 Bluetooth, BTS7960 motor driver, PZEM-004T, LCD/OLED displays, motor and pump control.

### Programming and Platforms:

C++, Embedded C, Arduino programming, ESP32 web-server-based monitoring, basic IoT dashboard development.

### Research and Documentation Tools:

MS Word, MS Excel, MS PowerPoint, Mendeley, EndNote, Publish or Perish 8, Canva, Grammarly.

### Practical Engineering Skills:

Circuit design, hardware prototyping, sensor interfacing, relay logic, load control, wiring, testing, troubleshooting, fabrication support, technical report writing, poster preparation, and presentation.

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## CO-CURRICULAR ACTIVITIES

- Delegate representing Netherlands in SPECPOL, MUET Model United Nations 2025.
  - Participant, MUET Model United Nations 2023.
  - Participant, Electrical Department Sports Week 2024, MUET Jamshoro.
  - Participant, MASS Masstival 2.0, Mehran Architecture Students' Society, 2024.
  - Completed 3-week Gurukul Training Course at ISKCON Hyderabad Sindh, 2024.
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## LANGUAGES

- English – Professional Working Proficiency
- Urdu – Native / Fluent
- Sindhi – Native / Fluent